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ME3330 – Thermal Sensors Technical Summary

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Deliverable 1:

Sensitivity is a measurement of how much a sensor will react to changes in its environment. This is important because a sensor with higher sensitivity is capable of reacting more quickly to a changing environment. This can be important in rapidly changing environments, or environments where data is infrequent, and a less sensitive sensor might not detect such a quick event. Thermal sensor sensitivity is measured in units of $[\frac{V}{°C}]$, and its formula is seen below as Equation 1.

$$S = [\frac{\Delta output}{\Delta \text{input}}] \quad (1)$$

Another characteristic of sensors is resolution. Sensor resolution is the smallest change in an input that can be detected by a sensor. Note that this is a small change in measured value, and not to be confused with a short time interval. Resolution uses units of Volts, and the process to calculate theoretical resolution can be seen below in Equation 2

$$Q[V] = \frac{V_{range}}{2^n} \quad (2)$$

Equations 1 and 2 were used alongside experimental data to fill out table 1

Sensors	Sensitivity [V/°C]	Experimental Resolution [°C]	Theoretical Resolution [°C]
TR	0.052	0.500	0.004
TC	0.089	0.390	0.1501
USB-TC	0.177	0.78	0.003

TABLE 1: This table contains sensitivity and resolution from experimental and calculated values for comparison.

The theoretical resolution of all sensors was lower than the experimental results. This can be seen above in TABLE 1. The relative difference between the resolution of the three sensors theoretical values were large, and they are relatively close together in the results from the experiment. This suggests that the cause of the discrepancy affected all the sensors. One possibility is the DAQ. The experimental resolution was found by finding the smallest slope between any two recorded data points. The time interval used was 0.25s and this may have given the sensors more time between data points than their theoretical resolution accounts for. This time interval could be reduced, and the resolution would increase.

Deliverable 2:

During the experiment, a transient test was performed where the sensors were taken from a steady state cold value and moved to a steady state hot value. The data recorded throughout the test can be seen in FIG. 1 below.

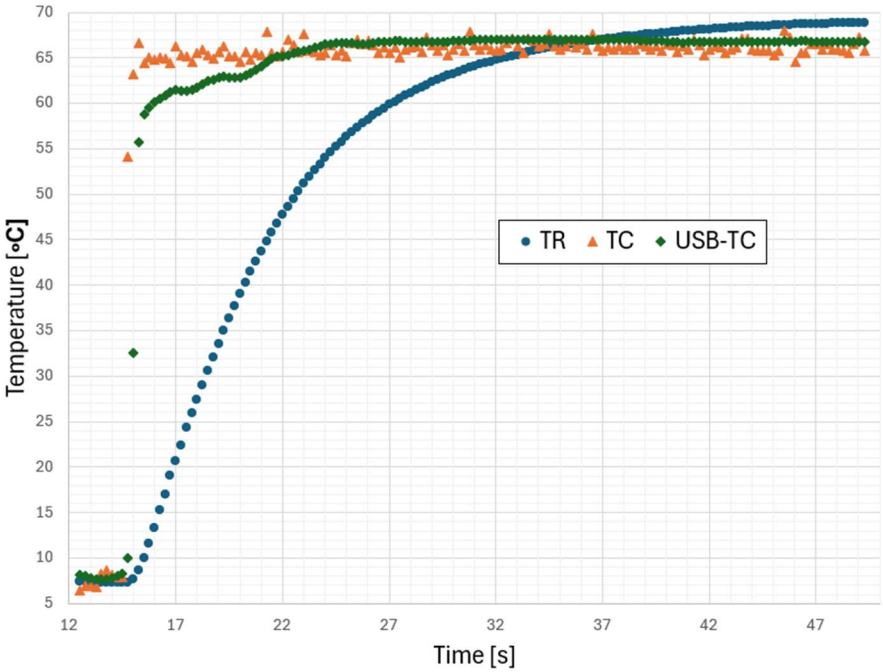


FIGURE 1. Experimental results from transient test for all three sensors

The time constant is defined as the time it takes for a sensor to reach 63% of the new steady state value. This is important because it can be used for both the selection and comparison of a sensor. A sensor that can quickly reach its new steady state could be useful in some situations. It is important to note that sensitivity is a measure of how voltage values per change in temperature, and the time constant measures the time taken to reach 63% of the new steady state temperature. A sensor with a high sensitivity can have a short time constant, and a sensor with low sensitivity does not necessarily have a short time constant. The time constant can be found by solving for T_τ below in Equation 3. T_τ is the temperature where the time constant occurs.

$$T_\tau = T_C + 0.63\Delta T \tag{3}$$

After solving for the time constant temperature, linear interpolation is used to estimate the exact time constant. This is required due to the high sensitivity of TC, and USB-TC which leave few values in the transient part of the plot between the steady states. The linear interpolation is completed using Equation 4.

$$t_\tau = t_C + \frac{(t_H - t_C)}{(T_H - T_C)}(T_\tau - T_C) \tag{4}$$

The value for time constant is now in terms of the final equilibrium point. This value will be normalized by subtracting the starting time at steady state cold to get an absolute value that is independent of this graph. This is shown below in Equation 5.

$$\tau = t_\tau - t_{ss,c} \tag{5}$$

With the value centered around zero, these time constants can accurately compare the sensors.

Sensor Type	T_τ [℃]	t_τ [s]	τ [s]
TR	46.345	21.635	10.631
TC	44.499	14.704	3.699
USB-TC	45.049	15.141	4.137

TABLE 1: This table contains the temperature constant, the time constant before adjustment, and the time constant after adjustment.

The three sensors all have varying time constants. The two thermocouples are over 1.5s apart, and the thermistor is almost 7 seconds behind the quicker thermocouple. Time constants are an important factor, but in terms of this experiment, the thermocouples quick time constants lead to them skipping over much of the data. Their quick jump is juxtaposed by the slower thermistor which was able to plot much more data through the transition phase. The role of the time constant in deciding what sensor to choose for a purpose is situational. If the data in between the equilibrium is valuable, then the thermistor is the better choice, but if the sensor needs to spend as little time as possible reaching the next equilibrium, then the thermocouples are the option to pick.

Deliverable 3:

Parameters for performance have been discussed in the previous sections, these include sensitivity, resolution, and time constants. This section will continue the discussion and comparison between the three sensors using noise level, cost, and temperature range.

The thermistor measures the difference in temperature by measuring resistance. This resistance value is relatively large, and due to this, the small wavelengths of noise have little effect on the final readings. This is opposed by the thermocouple. The thermocouple measures very small voltage changes, but these small values are susceptible to noise. The small values that noise occupies are hard to isolate due to the comparatively small values the thermocouple measures. Finally, the USB-Thermocouple falls in between the others. Still recording the small values, the USB amplifies the values which helps to isolate some of the noise. This leaves the USB thermocouple less effected than the thermocouple, but more susceptible than the thermistor.

Cost is a large factor when considering implementation into large scale production. The USB-thermocouple is by far the most expensive, due to its requirement of needing a USB interface. These can become expensive depending on the complexity required. The regular thermocouple can be had for much cheaper due to losing the complexity of the USB. The cost remains up due to the need for signal amplification. Finally, the thermistor is the cheapest due to not requiring anything to supplement its operation.

Temperature Range is an important consideration for a thermal sensor. The applications where thermal sensors are required are usually those of extreme temperatures, and therefore a comparison of the windows of operation is crucial. The thermistor is cited to operate from -100 to 325℃, and the type t thermocouple is cited to operate from -270 to 370℃. The thermocouples temperature range completely encompasses the thermistor range. If temperature range is a large consideration, then the thermocouple will prove more versatile than the thermistor.

Sensor	Noise Level	Cost	Temperature Range
TR	Lowest	Lowest	Lowest
TC	Highest	Moderate	Highest
USB-TC	Moderate	Highest	Highest

TABLE 3: Summary of points made throughout deliverable 3.

Deliverable 4:

Each sensor will be evaluated by how long it takes to obtain a 95% certainty for a confidence interval of 0.1[℃] at the hot steady state temperature. This process is an iterative one that requires usage of uncertainty analysis to derive the number of measurements required. The equation for the number of measurements is listed below as Equation 6

$$N = \left(\frac{t_{v,P} * S_x}{\frac{CI}{2}}\right)^2 \tag{6}$$

The $t_{v,P}$ is what is obtained from the uncertainty analysis, the S_x is the standard deviation of the hot steady state, and CI is the confidence interval. The issue that arises when solving is that $t_{v,P}$ is dependent on the degrees of freedom v. The equation for v is listed below as Equation 7

$$v = N - 1 \tag{7}$$

When solving for v and N, there are three possible cases. Case one is that $v < 0$, which leads to N = 1. Case two is when N jumps between two values, in which the larger value is chosen. Finally, if N > 61 then the number of measurements is uncertain. The iterative process for case one will be shown below.

Iteration 1: USB-TC

$$S_x = 0.0346, N = 61$$
$$v = N - 1 \tag{7}$$
$$v = 60$$

From the uncertainty analysis, at $v = 60$, $t_{v,P} = 2$

$$N = \left(\frac{t_{v,P} * S_x}{\frac{CI}{2}}\right)^2 \tag{6}$$
$$N = 0.4667$$
$$v = N - 1 \tag{7}$$
$$v = -0.5333$$

As can be seen, v < 0, therefore this suits case one. The remaining sensor iteration values can be seen below in TABLE 4.

Sensor	Case	Number of required Measurements
TR	2	2
TC	3	Uncertain
USB-TC	1	1

TABLE 4: Results of iterative process detailed above

REFERENCES

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